HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 16 Name: _____ Marks: / 54

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

   (a) Write down the value of p. (1 mark)

   (b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

   (i) Find q and r.

   (ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. The coordinates of the points Q and R are $(10, -1)$ and $(-4, -9)$ respectively.

(a) Let P be a moving point in the rectangular coordinate plane such that $PQ = PR$. Denote the locus of P by Γ.

(i) Describe the geometric relationship between Γ and QR.

(ii) Find the equation of Γ. (3 marks)

(b) Let C be the circle which passes through Q, R and the point $(4, 3)$.

(i) Find the equation of C.

(ii) The coordinates of the point U are $(10, 4)$. It is found that U lies outside C. UV and UW are the tangents to C at the points V and W respectively. Is the area of the circumcircle of $\triangle UVW$ greater than 100 ? Explain your answer. (5 marks)

(8 marks) (DSE-2024 PI Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

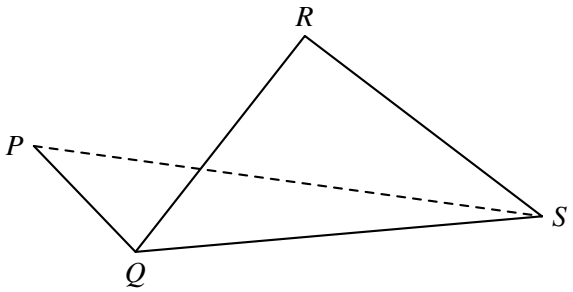


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. Let m be a constant. Solve the equation $x^2 - 3x = (m-1)^2 - 3(m-1)$.

A. $x = m - 1$ or $x = m - 4$

B. $x = m - 1$ or $x = 4 - m$

C. $x = 1 - m$ or $x = m - 4$

D. $x = 1 - m$ or $x = 4 - m$

(2 marks) (DSE-2024 PII Q07)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. Find the constant k such that the straight lines $6x - 8y = 7k$ and $kx + 12y = 5$ do not intersect with each other.

A. -16

B. -9

C. 9

D. 16

(2 marks) (DSE-2024 PII Q26)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188.$

B. $192 \times 16^{11} + 205.$

C. $175 \times 16^{12} + 188.$

D. $192 \times 16^{12} + 205.$

(2 marks) (DSE-2024 PII Q32)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. Let $z=(a-5)i+\dfrac{(a+2)i}{2+i}$. If a and z are real numbers, then $a-z=$

A. 2.

B. 3.

C. 4.

D. 5.

(2 marks) (DSE-2024 PII Q35)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. Consider the following system of inequalities:

$$\begin{cases} x - 2y \leq 1 \\ x + 4y \leq 13 \\ 2x - y \geq -1 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. Find the constant c such that the least value of $5x - 2y + c$ is 22, where (x, y) is a point lying in R.

A. 1

B. 23

C. 25

D. 29

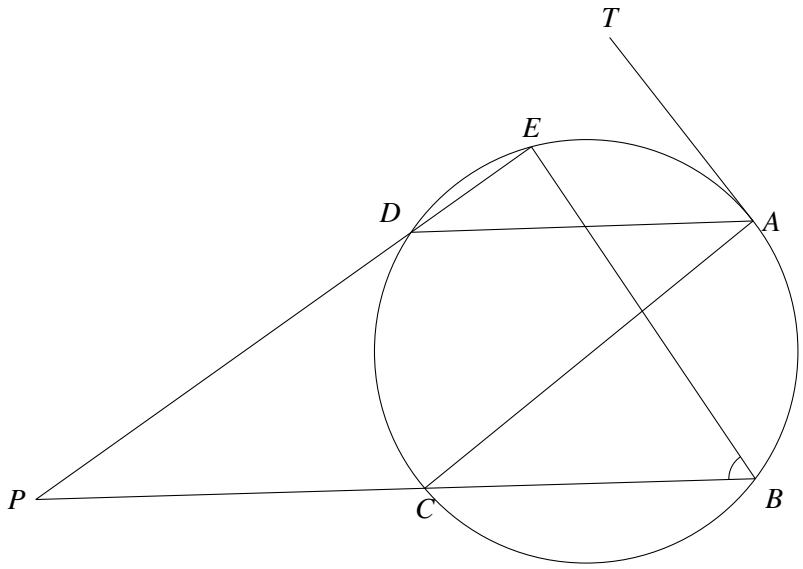
(2 marks) (DSE-2024 PII Q37)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. In the figure, TA is the tangent to the circle $ABCDE$ at the point A. BC produced and ED produced meet at the point P. If $\angle ACB = 43°$, $\angle DAT = 55°$ and $\angle CPD = 29°$, then $\angle CBE =$



A. 64°.

B. 69°.

C. 72°.

D. 78°.

(2 marks) (DSE-2024 PII Q38)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. There are three questions in a test. The probabilities that a student answers the first question correctly, the second question correctly and the third question correctly are 0.6, 0.7 and 0.8 respectively. Find the probability that the student answers at least 1 question correctly in this test.

A. 0.024

B. 0.188

C. 0.812

D. 0.976

(2 marks) (DSE-2024 PII Q43)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. If the variance of the seven numbers x_1, x_2, x_3, x_4, x_5, x_6 and x_7 is 16, then the standard deviation of the seven numbers $9x_1 - 5$, $9x_2 - 5$, $9x_3 - 5$, $9x_4 - 5$, $9x_5 - 5$, $9x_6 - 5$ and $9x_7 - 5$ is

A. 31.

B. 36.

C. 139.

D. 144.

(2 marks) (DSE-2024 PII Q45)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.